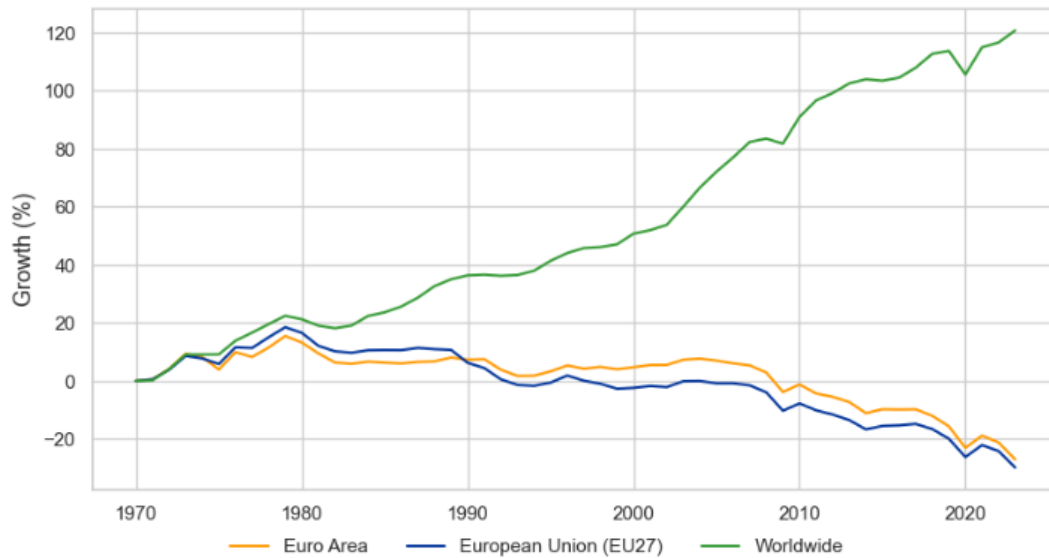
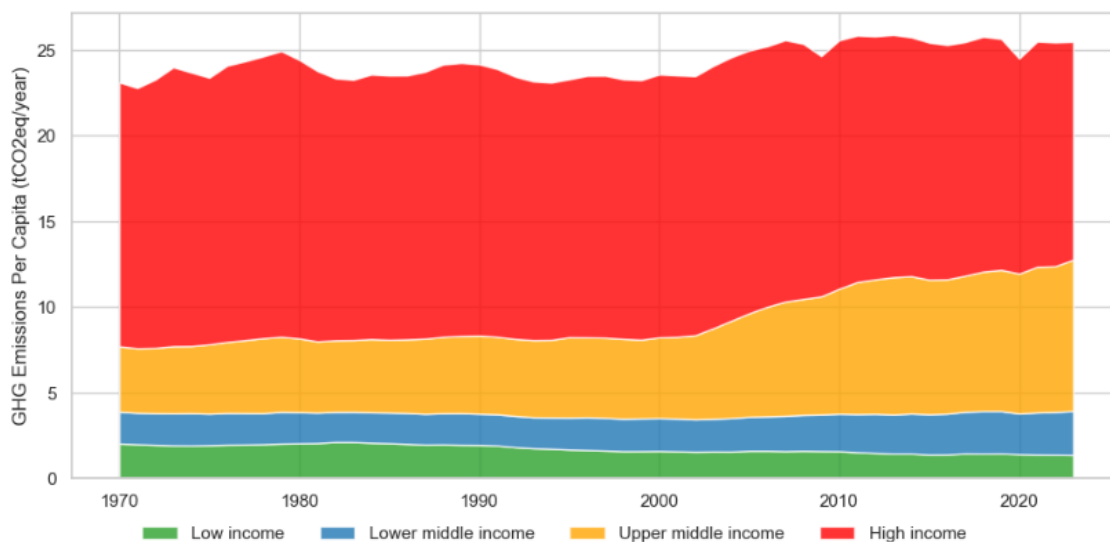


**Chart 1 – Evolution of GHG Emission Growth in the Euro Area, European Union (EU27), and Globally (Baseline Year: 1970)**



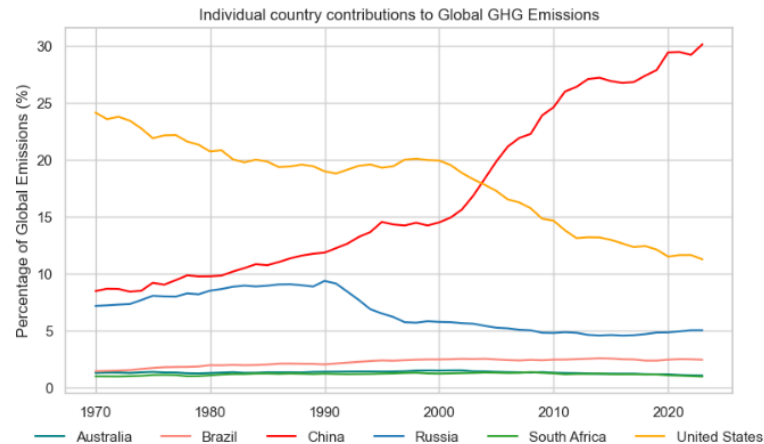
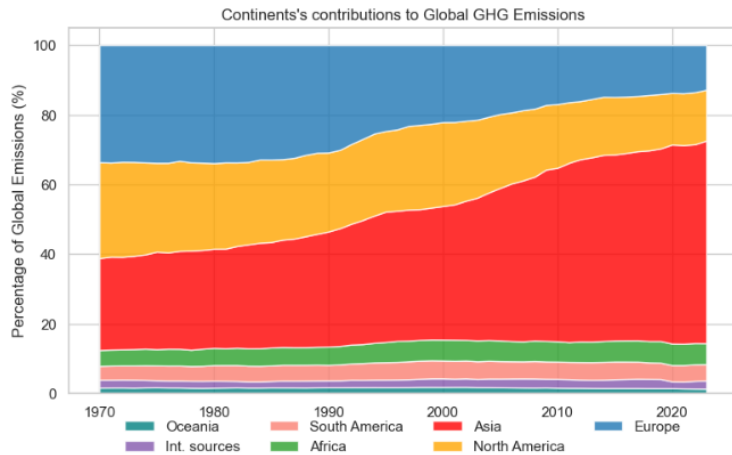
- Since 1970, **global GHG emissions** have steadily increased reaching a **120% rise by 2023**, reflecting ongoing industrialization and economic growth, particularly outside the EU27.
- **EU27 and Euro Area GHG emissions** peaked in the late 1970s and have since declined significantly. By 2023, emissions in the EU27 and Euro Area **decreased by 30% and 27%**, respectively, from their levels in 1970.
- While global emissions continued to increase, the **Euro Area and EU27** saw their growth rates turn negative in the late 2000s. This contrast highlights their **sustained efforts to reduce emissions**.

**Chart 2 – Comparison of countries' GHG emissions per capita aggregated according to the World Bank income groups.**



- **High-income** countries consistently recorded the highest GHG emissions per capita, but since the early 2000s, levels declined **from about 15 to 12.7 t CO2eq/year by 2023**.
- **Upper-middle-income** countries experienced a significant rise in per capita emissions, increasing **from around 4.7 in 2000 to 8.8 t CO2eq/year in 2023**, indicating rapid industrialization and economic growth.
- **Low-income** countries maintained relatively stable per capita emissions decreasing slightly **from 2.04 to 1.37 t CO2eq/year**, while **lower-middle-income countries** showed modest increases **from 1.86 to 2.57 t CO2eq/year**, indicating slower economic development compared to their upper counterparts.

**Chart 3 – Contribution of individual country and continent GHG emissions to total world GHG emissions.**



- **Europe** reduced its emission share **from about 34% to 13%**, and **North America from 24% to 11%**. This reflects effective emissions reduction policies, particularly evident in the declining contributions from Russia and the U.S.
- **Asia** has emerged as the leading contributor to global GHG emissions, with its share rising **from 26.4% in 1970 to over 58% by 2023**. This growth is largely driven by rapid industrialization and economic expansion across the region, especially in China.
- Contributions from **Africa, South America, international sources, and Oceania** remained relatively stable over this period, with minor fluctuations. Together, they **account for less than 15%** of total global emissions.

## Methodology

The data underlying this analysis was sourced from the GHG\_totals\_by\_country dataset in the EDGAR GHG emissions database. The downloaded dataset was imported into Python to enable efficient preprocessing, manipulation, and visualization. Pandas and NumPy libraries were used for data cleaning and transformation, while Seaborn and Matplotlib were used to produce the graphs.

The dataset reports GHG emissions expressed in megatonnes of CO2 equivalent per year (Mt CO2eq/yr) for each country, as well as aggregates for the EU27 and globally, from 1970 to 2023.

A **line chart** was used in Chart 1 for its simplicity and effectiveness in showing the evolution of GHG growth over time in the Euro Area, EU27, and globally. Data for the Euro Area was computed by subtracting the values of non-Euro Area countries ('BGR', 'CZE', 'DNK', 'HUN', 'POL', 'ROU', 'SWE') from the EU27 aggregate. The growth rate was calculated as a percentage relative to the baseline year of 1970 for each area. This approach was preferred as it highlights the cumulative growth from a consistent starting point, offering a clearer view of long-term developments.

A **stacked area chart** was used for Chart 2 to compare GHG emissions per capita across World Bank income groups. This chart type was chosen because it highlights differences among groups while preserving the temporal dimension, priority was given to the overall trend rather than a discrete comparison at a specific snapshot. The income group classifications and population data for each group were obtained from the World Bank's official website, and each country was mapped to the correct group using its country code. GHG emissions per capita were calculated by dividing the total emissions of each group by its population, and values were then expressed in tonnes of CO2 equivalent per year (t CO2eq/yr) to enhance readability. This approach was preferred over averaging, for each group, the GHG emissions per capita available in the EDGAR dataset, as it accounts for the relative weight of each country and provides a more accurate representation of reality.

A **stacked area chart** and a **line chart** were used for Chart 3 to represent the contribution of continents and individual countries to global emissions. A stacked area chart was chosen because it illustrates the relative contribution of each continent while preserving the temporal dimension, as before, priority was given to the overall trend rather than a discrete comparison at a specific snapshot. Countries were mapped to each continent using their country code and the Pycountry library in Python. Total emissions by continent were then divided by the global emission. International sources were also displayed to give a correct representation of the global emissions, although not directly allocable to one continent. For simplicity we picked only the main producer of GHG emission per continent, a line chart was in this case preferred to better highlight the individual trend of each country and explain the general development of the respective continent rather than comparing the relative contributions of single countries.